Bring Them Home

Flight Director · Lunar Return Mission

15 SHIFTS · 52 STOPS · GRADE 12 · BRING_THEM_HOME

THE OPENING CARD

An explosion has crippled a ship coming home from the Moon. Three people are five days from Earth. Power is low, the cabin is cold, and the ship is drifting off its safe path. You are the flight director, which means every call the room makes is signed by you. Build the Return Checklist: one procedure a shift, and three people get home only if each one holds.

The Failure

BEFORE THE SHIFT — GREET

Meet the six people who can save the crew

Meet each lead now so you know who owns the ship's path, power, air, radio, motion, and final calls.

PLAN CARD 40W

Shift 1. An explosion hit the ship four minutes ago. Power is falling, and three pressure screens show the same sharp drop. You must find whether the cabin is leaking before a wrong order wastes air, power, or crew time.

BRIEFING 18W

An explosion has damaged the ship and its sensors disagree. Decide what is real before the crew acts.

WHAT YOU DECIDE 14W

Create a trusted state vector and a prioritized anomaly list before commanding the crew.

1.1 What failed: the spacecraft or the sensors?

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · TRACE

SCENE 39W

Nine minutes after the bang, three digital cabin-pressure channels fall together. A mechanical gauge stays steady, and the acoustic leak monitor hears no escaping gas. Carter, the mission integration lead, needs one call before the crew opens a panel.

GUIDE 58W

Open each pressure channel to see what it depends on. Keep the ones that would still stand if the sensor electronics were at fault, and untick the rest. Then name the shared source that moved the others. Three channels falling together is not three measurements agreeing; it is one measurement repeated three times if they share a supply.

ASK 14W

Which pressure evidence survives the dependency check, and what shared source moved the rest?

BEHIND THE BUTTON 155W

Why three digital channels can be one measurement. Redundant transducers usually share a power supply, a reference voltage and a signal conditioning card. Any of those failing moves all three readings the same way at the same moment, which is exactly what a real depressurisation looks like on the display.

What the other two instruments are for. A mechanical gauge has no electronics between the cabin and the needle. An acoustic monitor listens for escaping gas rather than measuring pressure at all. Neither depends on the suspect chain, which is why they carry more weight here than any number of digital channels.

Why the diagnosis has to come before the panels. Opening panels in a cabin that is genuinely losing air spends time the crew may not have. Sending them to chase an instrumentation fault costs the same hour with no leak to find. The flight director needs one answer, not a list of possibilities.

TAKES AS READ

several instruments can share one reference circuit · an independent measurement does not inherit that shared failure

VERDICT 74W

Agreement is strong evidence only when the measurements can fail independently. The three digital pressure channels all inherit the same reference circuit. When that reference shifts, all three move together even if cabin pressure does not. The mechanical gauge has a different physical path, so its steady reading still counts. The acoustic monitor also finds no leak. Tracing dependencies turns three alarming numbers into one electrical fault and preserves the evidence that still stands.

TAKEAWAY 16W

Several agreeing readouts count as one piece of evidence when they inherit the same physical reference.

1.2 Build the first state estimate

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · SEQUENCE

SCENE 26W

One power bus reads zero. The guidance unit has lost its record. Carter asks you to rebuild the log so the next shift can trust it.

GUIDE 64W

All four of these will be done, so this is not a list of options. Ask of each what would be lost or wasted if it came later. Telemetry across the failure is overwritten within minutes. A fit across mixed clocks produces a confident answer that is wrong. And a command is only worth issuing once the estimate it rests on carries an uncertainty.

ASK 14W

Put the four cards in order. Start with the step that must happen first.

BEHIND THE BUTTON 169W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards have nothing before or after them. That constraint usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a number without its units and its time is not a measurement

VERDICT 83W

A state estimate is only worth having if it survives being wrong, and the order is what makes that possible. Telemetry across the failure is overwritten within minutes, so it is frozen before anything is computed from it. Units, timestamps and calibration come next, because a fit across mixed clocks produces a confident answer that is wrong. Estimates then carry uncertainties rather than single values. And the commands that follow are the ones whose benefit holds across every state the data still allows.

TAKEAWAY 12W

A good emergency command works across uncertainty rather than assuming it away.

1.3 Spend the first ten minutes

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · CHOICE

SCENE 27W

Four teams want the crew's radio at once. The next tracking pass starts in ten minutes. You must set one order before mixed commands reach the ship.

GUIDE 59W

All four options are real work that has to happen in the next hour. Ask of each what it costs in the only resource that is short, which is attention. Four controllers are already working one failure from four sets of numbers. Two versions of the same analysis can reach the crew, and they will fly whichever arrives first.

ASK 14W

Ten minutes to the next tracking pass. What does the flight director do first?

BEHIND THE BUTTON 201W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Here it ran to 88 per cent of passage quizzes, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason looks exactly like a sound one. It stops looking like one on the day the approximation is what is being tested.

TAKES AS READ

two rooms acting on different versions of one number is worse than either acting alone

VERDICT 78W

Attention is a physical resource in an emergency, just like propellant. The first decision determines where that resource goes. Four controllers are already working one failure from four sets of numbers. Without one log and one command authority, the same analysis can be repeated or contradicted. The crew may then execute whichever version reaches them first. Organising the room is not a delay before the technical work. It is what lets every later calculation and command count once.

TAKEAWAY 11W

Attention is a scarce physical resource during a fast-moving systems failure.

END OF THE SHIFT 17W

Use units, vectors, graphs, measurement uncertainty, common-mode failure as an evidence chain rather than as isolated facts.

Find the Spacecraft

PLAN CARD 42W

Shift 2. Nine hours have passed. The ground still cannot place the ship. Reyes, the guidance lead, needs its position, while Whitaker, the trajectory officer, needs a path. You must join the last readings before the best tracking site drops from view.

BRIEFING 16W

The ship's computer lost its position and speed. Rebuild both from the tracking data that remain.

WHAT YOU DECIDE 13W

Produce a consistent trajectory estimate and identify which new observation most reduces uncertainty.

2.1 Which measurement constrains what?

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · PROTOCOL

SCENE 26W

The ship's computer lost its state, and tracking comes and goes. Each tool measures a different part of motion. Choose what the next pass must collect.

GUIDE 67W

Four things the state estimate needs, and four ways of measuring. Pair them by asking what each measurement is physically sensitive to. An angle? A travel time? A frequency shift? Or a difference between two earlier answers? One of the four is built out of the others, which is why it is the noisiest. The next pass is intermittent, so what gets scheduled has to be chosen.

ASK 13W

Match each quantity the state estimate needs to the measurement that provides it.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation. Choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced. The fault is usually not in the join you are struggling with. It is in one you made early and stopped questioning.

TAKES AS READ

describing where something is and how it is moving takes six numbers · waves: frequency, wavelength, speed — taken as read

VERDICT 79W

Each measurement constrains a different part of the spacecraft state. Direction comes from an optical angle; range comes from radio travel time; Doppler constrains line-of-sight speed. For a one-way 2 GHz carrier, 1 km/s of radial speed produces about a 6.7 kHz shift. A coherent two-way tracking link carries roughly twice the first-order Doppler because the signal experiences the motion on the uplink and downlink. Repeated positions can supply transverse motion and acceleration, but differencing measurements amplifies their noise.

TAKEAWAY 7W

Navigation is a reconstruction from partial projections.

2.2 Reconstruct the state vector

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A PERSON · SEQUENCE

SCENE 23W

One range, two angles, and one Doppler trace remain. Their clocks do not match. Build a state that can predict the next pass.

GUIDE 63W

All four steps will happen, so ask what each one needs to be true already. A fit across two clocks can look clean and still be wrong, so the frames come first. Range, angle and Doppler constrain different parts of the motion and are fitted together. Residuals separate a missing force from noise. And the burn is committed at an hour, not now.

ASK 23W

Put the four cards in order. The state vector comes out at the end, so start with the step that must happen first.

BEHIND THE BUTTON 169W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards have nothing before or after them. That constraint usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

measurements only combine when they share a frame and a clock

VERDICT 78W

A useful state estimate has to predict the next observation. First, every measurement is moved onto one coordinate system and one clock. Otherwise a clean fit can still be wrong. Range, angle, and Doppler are then fitted together because they constrain different parts of the motion. The residuals show what the model missed. A pattern suggests an unmodeled force; random scatter suggests noise. Finally, both the state and its uncertainty are propagated to the time of the burn.

TAKEAWAY 12W

A useful state estimate predicts future observations and exposes its own failures.

2.3 What holds it in the curve

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A PERSON · CHOICE

SCENE 33W

The free-return trajectory is a lunar flyby, not a circular orbit. The velocity vector turns continuously under lunar gravity while the speed and the range both change. Explain why it misses the surface.

GUIDE 48W

Gravity is the external force here and it points toward the Moon. The spacecraft also arrives with sideways velocity. Ask which answer describes those two facts without inventing lift or an outward force. This is a flyby: the speed and direction keep changing as gravity bends the path.

ASK 14W

Why does the free-return path bend around the Moon instead of simply hitting it?

BEHIND THE BUTTON 201W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Here it ran to 88 per cent of passage quizzes, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason looks exactly like a sound one. It stops looking like one on the day the approximation is what is being tested.

TAKES AS READ

gravity continuously changes a spacecraft's velocity vector during a flyby · kinematics — taken as read · newton's laws and free-body reasoning — taken as read

VERDICT 92W

The spacecraft is in continuous free fall, but free fall does not mean falling straight down. Lunar gravity accelerates the vehicle toward the Moon while its existing sideways velocity carries it across the face of the Moon. The combined motion bends the trajectory around the body. On a free-return the path is generally not circular: speed and distance from the Moon change through the encounter. A trajectory with too little transverse speed or the wrong aim can intersect the surface; a suitable incoming state swings past and departs on a new direction.

TAKEAWAY 16W

> A gravity assist or free-return flyby is continuous free fall on a curved, generally noncircular trajectory.

END OF THE SHIFT 18W

Use position, velocity, acceleration, coordinate systems, graphs of motion as an evidence chain rather than as isolated facts.

SHIFT 3 OF 15

The Wrong Trajectory

PLAN CARD 49W

Shift 3. The drift is real, and the current path misses safe entry by about 400 kilometres. A small burn now may fix it, but a burn based on bad data would make things worse. You must decide if the miss is large and clear enough to spend fuel.

BRIEFING 20W

The ship will miss the safe path to Earth. Decide if the evidence is strong enough to burn fuel now.

WHAT YOU DECIDE 14W

Choose a correction that restores a safe return while preserving fuel and engine options.

3.1 From force to trajectory change

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · SEQUENCE

SCENE 28W

The ship is a little off course now, but the miss grows with time. The crew asks why a short burn now can replace a long burn later.

GUIDE 59W

These four are one chain, so each card has something before it and something after. Ask what has to exist first. A force before an acceleration. An acceleration before a change in velocity. The last card is why any of it matters. The changed velocity is carried for hours before arrival, which is why seconds now beat minutes tomorrow.

ASK 20W

Put the four cards in order. Each card is one link from the engine thrust to the trajectory it changes.

BEHIND THE BUTTON 169W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards have nothing before or after them. That constraint usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a force acting on a mass produces an acceleration · kinematics — taken as read

VERDICT 78W

The chain starts with force applied for a measured time. Force divided by spacecraft mass gives acceleration. That acceleration changes velocity during the burn. The impulse, force times time, equals the change in momentum. For a fixed spacecraft mass, that sets the change in speed. The new velocity then changes where the spacecraft will be hours later. This is why an early correction can be small: the vehicle carries the changed velocity for a long time before arrival.

TAKEAWAY 11W

Trajectory control acts on velocity now to change position much later.

3.2 Small burn, large consequence

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A PERSON · BALLPARK

SCENE 22W

A 30,000 kg ship gets 6,000 N of thrust for 20 seconds. Find its change in speed before the burn is approved.

GUIDE 61W

Five numbers, and two of them belong to other questions. Surface gravity and specific impulse are not in this chain. Ask of each whether it is a force, a mass, or a time. And note the scale of the answer. Four metres per second looks like nothing beside an orbital speed. Days out, it moves the arrival by hundreds of kilometres.

ASK 5W

Estimate the change in speed.

BEHIND THE BUTTON 111W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before the tile is placed. It is also how you catch a quantity that belongs to a different part of the problem. That habit is what this format exists to build.

TAKES AS READ

force applied over a time is an impulse · newton's laws and free-body reasoning — taken as read

VERDICT 79W

Force divided by mass gives an acceleration of 0.20 metres per second squared. Over 20 seconds, that produces a 4 metre-per-second change in speed. The same result comes from impulse divided by mass. Four metres per second looks tiny beside an orbital speed measured in kilometres per second. Applied days before arrival, however, it shifts the encounter by hundreds of kilometres. A late correction must produce the same positional change in far less time, so it costs much more.

TAKEAWAY 10W

Orbital corrections often rely on small velocity changes applied early.

3.3 What the pressure difference is for

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · CHOICE

SCENE 32W

The oxygen tank is at 40 psi, and the regulator is set to 15 psi. Yet the flow has fallen by half. Brooks, the life support lead, asks what could block it.

GUIDE 62W

All four options explain a flow that has halved. They differ in what they hold responsible: the difference, the path, the sensor, or the temperature. Ask of each whether the line is the same line it was yesterday. A pressure difference drives the flow, and how restrictive the path is decides the rate. A high-pressure tank delivers nothing through a blocked line.

ASK 15W

The tank reads 40 psi and the regulator wants 15. What is the flow doing?

BEHIND THE BUTTON 201W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

gas flows from higher pressure to lower pressure · newton's laws and free-body reasoning — taken as read

VERDICT 82W

Gas moves when a pressure difference exists across a path. The flow rate also depends on how restrictive that path is. A 40 psi tank feeding a 15 psi regulator still has a strong driving difference. If the measured flow falls by half, either that difference changed or resistance in the path increased. A partly closed valve, ice, or a loaded filter can do that. Tank pressure alone never guarantees flow, because even a high-pressure tank delivers nothing through a blocked line.

TAKEAWAY 14W

Something moves through a system for one reason and is held back for another.

END OF THE SHIFT 17W

Use Newton’s laws, gravity, orbits, impulse, trajectory sensitivity as an evidence chain rather than as isolated facts.

Turn Without Wasting Fuel

BEFORE THE SHIFT — FOLLOW

Shadow the live handover

Follow the outgoing controller so no lost fact becomes the next shift's wrong order.

PLAN CARD 42W

Shift 4. The main guidance unit is off, and one thruster set is dead. The crew must turn the ship 90 degrees by hand and stop at the right angle. A bad turn could lose the radio link and the next burn.

BRIEFING 20W

The main guidance unit is off and one thruster set is dead. Write a hand-flown turn that stops on target.

WHAT YOU DECIDE 13W

Develop a controlled reorientation procedure that stops the rotation at the required attitude.

4.1 Read rotational motion

ROTATION & STRUCTURES · STRUCTURES TEST BAY · A A PLACE · PROTOCOL

SCENE 22W

The crew must fly the turn by hand. Mensah, the dynamics lead, wants the rules of rotation clear before any thruster fires.

GUIDE 67W

Four changes on the left, and four consequences on the right. Pair them by asking what each change does to the two terms that matter: the torque, and the resistance to turning. One of the four adds no external torque at all, which is what makes it different from the others. The crew is about to fly this by hand and will feel every one of them.

ASK 12W

Match each case to the best scientific answer. Use each choice once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation. Choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced. The fault is usually not in the join you are struggling with. It is in one you made early and stopped questioning.

TAKES AS READ

a torque is what makes something start or stop turning · kinematics — taken as read

VERDICT 82W

Attitude control is Newtonian mechanics about a centre of mass. Torque grows with force and with the lever arm from that centre. The same torque produces less angular acceleration when the moment of inertia is larger. Equal and opposite torques applied for equal times cancel the net angular impulse. Internal mass motion can change how fast different parts rotate, but it cannot change the vehicle's total angular momentum without an external torque. Those four rules set the logic of every manual turn.

TAKEAWAY 10W

Attitude control is Newtonian mechanics around a center of mass.

4.2 Execute a manual attitude maneuver

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · FLY

SCENE 35W

The guidance platform is off. The crew must turn the ship 90 degrees using two opposing thruster pairs and a window reticle. Mensah reminds the loop that every extra correction spends propellant reserved for entry.

GUIDE 55W

You set two things: how hard to start the turn and where to start braking. Then the plan runs and you watch it. Nothing damps this rotation, so whatever you start you must stop deliberately, and the braking has to lead the target rather than meet it. Every extra correction spends propellant reserved for entry.

ASK 19W

How hard do you start the turn, and where do you begin braking so the ship arrives at rest?

BEHIND THE BUTTON 153W

Why the brake has to lead. With nothing to slow the spacecraft, the rotation continues at whatever rate you gave it until an equal impulse takes it away. Firing at the target means arriving at the target still turning, and then correcting past it — which is where the propellant goes.

What the reticle actually gives you. A mark on the window and a star or a horizon is enough to see the attitude, not the rate. Rate has to be inferred from how fast the view moves. That is why a small starting impulse is easier to fly than a large one, even though it takes longer.

Why the propellant matters more than the minutes. What is being spent belongs to the entry budget. A turn flown in half the time with three corrections can cost the margin that the atmosphere entry needs, and there is no way to get it back.

TAKES AS READ

a torque changes angular speed · rotation continues after a thruster stops firing · newton's laws and free-body reasoning — taken as read · rotational kinematics — taken as read

VERDICT 81W

A thruster pulse changes angular speed; it does not create a turn that stops by itself. After the opening pulse ends, the spacecraft keeps rotating at the new rate. The opposite thruster needs time and angle to remove that rate. For equal start and stop pulses, the braking distance is the same kind of problem as stopping a moving cart. A late brake therefore overshoots. The maneuver is successful only when both angle and angular rate are small at the target.

TAKEAWAY 17W

In undamped rotation, stopping the torque does not stop the motion; braking must begin before the target.

4.3 Torque from a thruster

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A PERSON · BALLPARK

SCENE 24W

A 200 N thruster fires at a right angle, 3 metres from the ship's centre. Find the torque before the crew predicts the turn.

GUIDE 61W

Five numbers, and three of them belong elsewhere: the vehicle mass, the docking ring, and a cosine. Ask of each whether the turning effect depends on it. The force is at right angles here, so no angle factor is needed. And note what the crew can actually change. Choosing a different thruster pair changes the lever arm without changing the thruster.

ASK 8W

Estimate the torque about the centre of mass.

BEHIND THE BUTTON 111W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before the tile is placed. It is also how you catch a quantity that belongs to a different part of the problem. That habit is what this format exists to build.

TAKES AS READ

a longer lever arm gives the same force more turning effect · newton's laws and free-body reasoning — taken as read · rotational kinematics — taken as read

VERDICT 78W

Torque is force times lever arm times the sine of the angle between them. Here the force is perpendicular, so that sine is one. The calculation is therefore just 200 newtons times 3 metres. The result is 600 newton metres. Move the same thruster closer to the centre of mass and the torque falls in direct proportion. On this vehicle, the crew can choose which thruster pair fires. That choice changes the lever arm without changing the thruster.

TAKEAWAY 11W

Force location matters as much as force magnitude in rotational control.

4.4 Off-nominal, or off-scale

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · BELT

SCENE 31W

The screen has more alarms than the team can check. Some show a real change in the ship. Others show a bad sensor. Lindqvist, the flight dynamics integrator, needs them sorted.

GUIDE 50W

Two bins. A spacecraft doing something unexpected moves in ways that are physically possible and that other parameters agree with. A measurement failing moves in ways nothing else corroborates — instantly, off the end of the scale, or to exactly zero. Sort on whether the rest of the vehicle agrees.

ASK 16W

Send each flag to the bin that says whether the vehicle moved or the measurement failed.

BEHIND THE BUTTON 93W

Why this is the first question, always. The response to the spacecraft misbehaving and the response to an instrument misbehaving are opposite. One is an intervention. The other is ignoring a reading and fixing a sensor. Getting it backwards spends a consumable answering something that never happened.

What a failed sensor looks like. A step no physical process could produce. A value past the end of the range. A reading that has not changed in an hour when everything around it has. Or a number that agrees with nothing else on the vehicle.

TAKES AS READ

a reading comes from a sensor, and a sensor can fail

VERDICT 125W

Every flag on this board is a number that moved, and the number is not the thing. A spacecraft that is genuinely warming shows it in more than one place. The structure temperature moves with the radiator outlet, and the power draw agrees. A failed sensor shows none of that. It steps instantly, parks past the end of its range, or freezes while everything around it changes. So corroboration is the test rather than plausibility: a failed sensor can produce an entirely plausible number. And the cost of confusing the two is asymmetric. Answering a sensor failure with an intervention spends propellant on a problem that does not exist. Treating a real excursion as a bad reading is how a crew runs out of options.

TAKEAWAY 15W

The vehicle and the measurement fail differently, and only one of them needs an intervention.

END OF THE SHIFT 21W

Use torque, angular acceleration, moment of inertia, angular momentum, center of mass as an evidence chain rather than as isolated facts.

SHIFT 5 OF 15

The Power Budget

PLAN CARD 44W

Shift 5. Power must now last 62 more hours. Shah, the power systems lead, sees a four-volt drop and a hot joint while every system asks to stay on. You must find the loss and save enough power for air, heat, guidance, and entry.

BRIEFING 21W

The fuel cells are gone and the batteries must last to entry. Find the power loss and choose what stays on.

WHAT YOU DECIDE 13W

Build a load-shedding plan that preserves essential functions through the next critical maneuver.

5.1 How long can the battery last?

POWER & CIRCUITS · POWER & CIRCUITS BAY · A A PLACE · BALLPARK

SCENE 27W

The fuel cells are gone. Entry batteries are now carrying the ship days too soon. Shah needs a rough count of hours before the next power cut.

GUIDE 68W

Five numbers, and three of them belong to other questions: the bus voltage, the brief peak, and the time to the next burn. Ask of each whether it is an amount of energy or a rate of use. Energy divided by a rate leaves a time. And note what the answer assumes: a flat load, and every last watt-hour available. A cold battery gives less than its rating.

ASK 4W

Estimate the ideal endurance.

BEHIND THE BUTTON 111W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before the tile is placed. It is also how you catch a quantity that belongs to a different part of the problem. That habit is what this format exists to build.

TAKES AS READ

energy is a stock, and power is the rate it is drawn at · circuits: current, voltage, resistance, Ohm's law — taken as read

VERDICT 81W

Energy and power are different things, and the division between them is the whole calculation. An inventory in kilowatt-hours divided by a draw in kilowatts leaves hours. It is an ideal figure. It assumes the load stays flat and that every last watt-hour is available, and neither is true — a cold battery gives less than its rating. It is still the number every other decision this shift gets argued against, so it is worth having early and worth calling approximate.

TAKEAWAY 15W

Power is the rate of energy use; endurance depends on both energy inventory and load.

5.2 Why is bus voltage collapsing?

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · CHOICE

SCENE 30W

The main bus has fallen from 28 volts to 24. Total current is near the plan, but one joint is hot. Find the fault before more gear is shut off.

GUIDE 63W

Four options, and three readings have to be explained together. Generation is normal. Total load current is near plan. Something is hot. Ask of each option how many of those it covers. A failed array or a hidden load would move the current. A bad meter would not make anything warm. Equipment is about to be pulled off the bus on this answer.

ASK 10W

Which fault fits the voltage, current, generation and thermal evidence?

BEHIND THE BUTTON 201W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Here it ran to 88 per cent of passage quizzes, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason looks exactly like a sound one. It stops looking like one on the day the approximation is what is being tested.

TAKES AS READ

current through a resistance drops voltage and makes heat

VERDICT 82W

Voltage, current, and heat describe the same electrical fault from different angles. A resistive junction creates a voltage drop when current passes through it. The lost electrical power becomes heat at that junction. Here generation is normal and the total load current is near plan, so neither a failed array nor a hidden load fits. The hot connection and low downstream bus voltage point to resistance in the path. Measuring all three quantities localizes the fault instead of guessing from voltage alone.

TAKEAWAY 19W

Electrical faults are easier to localize when voltage, current, and heat are treated as parts of one energy picture.

5.3 Shed load without losing the mission

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · ALLOCATE

SCENE 21W

The battery holds 41 amp-hours. Air, heat, and guidance need fixed blocks of time. Work out how much radio time remains.

GUIDE 62W

Forty-one amp-hours, sixty-two hours to fly, and a current draw of 4.2 amps that does not fit. Watch the questions list rather than the loads: it shows what the spacecraft can still do under the current plan. Every hour you buy for one load is an hour you have taken from another, and at least one has to come down within minutes.

ASK 10W

How many hours of each load can the ship afford?

BEHIND THE BUTTON 135W

What an amp-hour buys. One amp for one hour. At 4.2 amps the whole reserve lasts under ten hours, against sixty-two to fly. So this is not a trimming exercise. The load has to fall by a factor of several and stay down.

Why the loads are not equal. Guidance and communications can be intermittent — used for a pass and switched off. Thermal control has a time constant of hours, so it can be cycled. Life support cannot be interrupted at all. That difference is what makes a plan possible.

Why minutes matter for the first cut. Every hour spent deciding is drawn at the current rate, so a slow decision has already spent part of what it was allocating. The first reduction buys the time to make the rest of the plan properly.

TAKES AS READ

amp-hours equal current multiplied by time · some loads can be duty-cycled while others must run continuously · electrical power and energy budgets — taken as read · circuits: current, voltage, resistance, Ohm's law — taken as read

VERDICT 88W

Amp-hours are current multiplied by time. Continuous life support costs 0.4 A × 62 h = 24.8 Ah. Six hours of thermal protection at 1.1 A costs 6.6 Ah, and four guidance hours at 1.8 A cost 7.2 Ah. Those required loads total 38.6 Ah, leaving 2.4 Ah. At 0.9 A, that buys only about 2.7 hours of high-bandwidth communications. The right plan is therefore not 'turn off the least important system'; it is schedule each load for the hours in which its physical function is actually needed.

TAKEAWAY 20W

A finite energy budget is a set of tradeoffs between load rate, runtime, and the mission conditions each load protects.

5.4 Which measurement constrains what? — Review

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CALLBACK · PROTOCOL

SCENE 24W

Whitaker has one station and a 20-minute pass. Each reading misses part of the ship's state. Name each blind part before the pass begins.

GUIDE 64W

Four observations, and four things none of them can see. Pair them by asking what the instrument is physically sensitive to, then take the complement. A frequency shift responds to motion along the beam. A travel time responds to distance along a line. An angle fixes the line and nothing on it. And one of the four is built by subtracting two earlier answers.

ASK 12W

Match each observation to the part of the state it leaves unmeasured.

BEHIND THE BUTTON 166W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation. Choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

Why a blind axis is worth scheduling against. A pass is short, and the station can only be pointed at one thing at a time. So the observation worth buying is the one whose blind axis is already covered by something else. Two measurements blind in the same direction cost twice and buy once.

Why the transverse direction is the hard one here. A Doppler track is the cheapest measurement on the list and the one the room trusts most. It responds only to the component of velocity along the beam. A spacecraft drifting sideways at a kilometre a second moves the carrier by nothing.

TAKES AS READ

a measurement responds to one component of the motion and not to the others · subtracting two measured numbers carries both of their errors into the answer · waves: frequency, wavelength, speed — taken as read

VERDICT 85W

Each observation responds to one part of the motion and is silent about the rest. So the useful question about a short pass is not which measurement is best. It is which axis is still uncovered. Doppler carries the line-of-sight component of velocity and nothing across it. Range fixes a distance and leaves the direction open. A bearing does the opposite. Differenced positions give a rate and carry both errors forward. Two observations blind in the same direction cost two passes and buy one answer.

TAKEAWAY 21W

> Every observation is blind along some axis, and a schedule is chosen against the blind axes rather than the strong ones.

END OF THE SHIFT 17W

Use current, voltage, resistance, electrical power, energy budget as an evidence chain rather than as isolated facts.

A Dangerous Battery Configuration

PLAN CARD 42W

Shift 6. Ferreira, the electrical load officer, found a spare battery. It may add 11 hours, but its voltage does not match the live bus. The new joint is warm. You must test the current before the battery starts a cabin fire.

BRIEFING 23W

A spare battery could add hours, but a bad link could start a fire. Test the setup before it is used in flight.

WHAT YOU DECIDE 13W

Approve only a configuration whose voltage, current sharing, and failure behavior are understood.

6.1 Series or parallel?

POWER & CIRCUITS · POWER & CIRCUITS BAY · A A PLACE · PROTOCOL

SCENE 23W

A spare battery may buy more time. Ferreira asks how each wiring plan changes voltage, current, and the effect of one bad cell.

GUIDE 61W

Four wiring changes, and four consequences. Pair them by asking what stays the same in that arrangement: the voltage across each part, or the current through it. Then ask what one fault does. Adding a module can take capability away, because a single open cell can break a whole series path. This is being bolted up to buy hours of endurance.

ASK 12W

Match each case to the best scientific answer. Use each choice once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation. Choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced. The fault is usually not in the join you are struggling with. It is in one you made early and stopped questioning.

TAKES AS READ

a battery has both a voltage and a current it can supply

VERDICT 82W

Circuit topology determines voltage, current sharing, and fault behavior. Equal cells in series add voltage while carrying the same current. One open cell can therefore break the whole series path. Equal cells in parallel keep about the same voltage and share the current. That arrangement can add useful capacity. A resistive parallel branch carries less than its share, forcing the other branches to carry more. Redundancy only helps when the connection and protection rules prevent one fault from defeating the whole arrangement.

TAKEAWAY 11W

Redundancy is useful only when connection and protection behavior are explicit.

6.2 Heating at a bad connection

POWER & CIRCUITS · POWER & CIRCUITS BAY · A PERSON · BALLPARK

SCENE 22W

A new battery joint has 0.05 ohms of resistance and carries 20 amps. Find its heat rate before it starts a fire.

GUIDE 71W

Five numbers, and three of them belong elsewhere: the bus voltage, the whole cable run, and half the current. Ask of each whether the heating at this joint depends on it. Note the square: at half the current the same joint gives off a quarter of the power. And note where it is. Twenty watts is nothing on a radiator and a great deal in one connector in a sealed cabin.

ASK 7W

Estimate the heating power at the connection.

BEHIND THE BUTTON 111W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before the tile is placed. It is also how you catch a quantity that belongs to a different part of the problem. That habit is what this format exists to build.

TAKES AS READ

a resistance carrying current turns electrical power into heat

VERDICT 79W

Resistive heating is current squared times resistance, so the current term dominates. The same connector at half the current would give off a quarter of the power. The resistance sounds like nothing written down. The wattage it makes at this current is what decides whether the junction runs warm or starts a fire. 20 watts is nothing spread over a radiator. It is a great deal in one connector, in a sealed cabin, with no air moving over it.

TAKEAWAY 13W

Small resistances matter when current is large because heating scales with current squared.

6.3 Qualify the emergency battery plan

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · VERIFY

SCENE 27W

The spare battery and live bus have different voltages. The cable can carry only 8 amps. Predict the first current, then test it through a safe limit.

GUIDE 58W

Lock what current you expect before anything is connected. Then make the connection and decide whether to spend what it costs to measure the electrical effect. The connector is rated for 8 amps, and the module reads 31 volts against a nominal 37. So the prediction is about whether this is safe as well as whether it helps.

ASK 16W

What current do you expect, and will you verify the electrical effect before qualifying the module?

BEHIND THE BUTTON 141W

Why the voltage matters more than it looks. A module at 31 volts against 37 nominal is either partly discharged or damaged, and those imply very different currents when it is connected to a healthy bus. The current is set by the difference between the two, divided by the resistance of everything between them.

What the connector rating means. Eight amps is what the hardware can carry continuously. Exceeding it inside a sealed cabin is not a component failure — it is heat, and possibly smoke, in the volume the crew is breathing.

Why measure rather than assume. The prediction can be right and the hardware still behave differently: a damaged cell, a high-resistance joint, a connector not fully seated. Spending the measurement is what converts a plan into a qualification, and not spending it leaves the crew relying on arithmetic.

TAKES AS READ

a voltage difference between connected batteries drives current · a current rating is a physical limit, not a suggestion

VERDICT 82W

Two batteries at different voltages exchange current as soon as a conductive path joins them. The connection can look mechanically perfect while the current is already above a connector rating. Protection limits the consequence of being wrong, but it does not tell you what happened. The electrical result still has to be measured. Here the test lead survives, yet the measured current is too high for the intended connector. That makes the direct configuration unacceptable even though the test itself completed normally.

TAKEAWAY 18W

A successful electrical action is not a verified outcome until the current and heating it produces are measured.

6.4 Reconstruct the state vector — Review

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CALLBACK · SEQUENCE

SCENE 23W

The guidance unit has reset again. Hand-read star angles remain, but their clocks differ. Agree on the axes and clock before using them.

GUIDE 65W

All four of these will be done, so ask what each one needs to be true already. A direction is only a direction once somebody has said which way the axes point. Components cannot be added until they are components on the same axes. A velocity is a difference between two positions. And the crew read attitude in their own frame, not in the room's.

ASK 14W

Arrange the four cards from the earliest prerequisite or cause to the latest result.

BEHIND THE BUTTON 154W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

What a hand-taken mark actually is. A sextant angle between a star and a landmark is a direction and a moment, and nothing else. It carries no distance, no frame and no clock of its own, which is why all three have to be supplied before any arithmetic is done to it.

Why the last step is a rotation rather than a calculation. The state is computed in the frame the ground works in, and the crew fly the vehicle in theirs. The same vector has different numbers in the two, so a state handed across without the rotation is right and unusable.

TAKES AS READ

a direction is only a set of numbers once the axes have been named · components add only when they are components on the same axes

VERDICT 83W

A vector is a set of numbers together with the axes they are measured against, and the numbers mean nothing without the axes. So the frame and the clock are settled before anything is resolved. Each sextant mark then becomes components on those shared axes, which is what makes them addable at all. Position comes from combining components with a range, and velocity from differencing two positions. The rotation into body axes is last because it changes every number while changing nothing physical.

TAKEAWAY 26W

A vector is numbers plus the axes they are measured against, and the axes have to be settled before any of the numbers can be added.

END OF THE SHIFT 20W

Use series circuits, parallel circuits, internal resistance, Joule heating, fault isolation as an evidence chain rather than as isolated facts.

The Cabin Is Cooling

PLAN CARD 38W

Shift 7. The power cuts have pushed the cabin near 4°C. Water now beads on panels, and the coldest battery is needed for entry. You must place the last heater time where cold can do the most harm.

BRIEFING 21W

The power cuts have made the cabin cold and wet. Protect the crew and the entry batteries with very little heat.

WHAT YOU DECIDE 14W

Create a thermal survival plan that protects crew, electronics, and batteries with minimal power.

7.1 How much does the cabin cool?

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · BALLPARK

SCENE 30W

With most systems powered down, the cabin loses about a kilowatt more heat than it makes. Find the three-hour temperature trend before any scarce heater energy is spent on it.

GUIDE 64W

Five numbers, and two of them are the same three hours written in different units. Another is the present cabin temperature, which the drop does not use. Ask of each whether the relationship needs it. And note what the answer means. The cabin cools slowly because of what it is made of, which buys hours now and costs the same energy to reheat later.

ASK 13W

Estimate the temperature drop over three hours at the cabin's net thermal loss.

BEHIND THE BUTTON 111W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before the tile is placed. It is also how you catch a quantity that belongs to a different part of the problem. That habit is what this format exists to build.

TAKES AS READ

it takes energy to warm a mass, and the same energy leaves when it cools

VERDICT 73W

A lumped estimate treats the cabin, equipment, and contents as one effective thermal mass of about 12 MJ/K. A 1 kW net loss for three hours removes 10.8 MJ, corresponding to about 0.9 K of average cooling. The estimate does not say every surface cools by the same amount: local batteries and walls can be colder. Thermal inertia buys time, while the local temperature measurements decide where insulation or heater power actually matters.

TAKEAWAY 12W

Thermal inertia can make temperature change slowly even when power is lost.

7.2 Choose the heat-transfer mechanism

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A PERSON · PROTOCOL

SCENE 33W

The cabin keeps getting colder, and the crew wants to act. What can be done depends on how the heat is actually leaving. Name the pathway for each part before choosing a fix.

GUIDE 55W

Four situations, and four ways heat moves. Pair them by asking what carries the energy: touching, moving air, or nothing at all. One of these needs no material medium. The pairing decides what a fix looks like, because insulation, circulation and reflective surfaces each act on a different pathway. Naming the wrong one buys nothing.

ASK 12W

Match each case to the best scientific answer. Use each choice once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation. Choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced. The fault is usually not in the join you are struggling with. It is in one you made early and stopped questioning.

TAKES AS READ

heat moves by contact, by moving fluid, or by radiation

VERDICT 81W

Each heat-transfer pathway suggests a different intervention. Direct contact between people transfers heat by conduction. Moving cabin air transports heat by convection. Infrared exchange between warm objects and a cold wall is radiation, which needs no material medium. A metal strut carrying heat into colder structure is conduction through a solid. Naming the pathway matters because insulation, circulation, reflective surfaces, and thermal isolation act on different mechanisms. A useful fix starts by identifying which pathway is actually carrying the energy away.

TAKEAWAY 9W

> Identifying the pathway reveals which intervention can reduce loss.

7.3 Keep the cabin survivable

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CHOICE

SCENE 28W

The cabin is near 4°C, and water is forming on the walls. There is not enough power to heat the whole ship. Protect the parts that matter most.

GUIDE 63W

All four options are ways to spend a fixed energy budget. Ask of each whether it slows the loss or replaces energy already gone. Insulation does the first and cannot do the second. Then ask where cold costs most. Cold batteries deliver less usable energy, so a thermal problem becomes a power problem. The crew is asleep in their suits at four degrees.

ASK 15W

Four degrees and falling, on a fixed energy budget. What do you spend it on?

BEHIND THE BUTTON 201W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Here it ran to 88 per cent of passage quizzes, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason looks exactly like a sound one. It stops looking like one on the day the approximation is what is being tested.

TAKES AS READ

slowing a loss and replacing it are different things · newton's laws and free-body reasoning — taken as read

VERDICT 78W

Insulation slows heat loss; it does not replace energy that has already left. That makes insulation valuable when the power budget is fixed. Put it around the crew zone and critical batteries, where cold has the largest consequence. Cold batteries also deliver less usable energy, so a thermal problem can become a power problem. Heating the whole vehicle spends scarce energy on empty volume and exterior surface. Thermal survival is therefore an allocation across space, time, and consequence.

TAKEAWAY 11W

Thermal survival is an allocation problem across space, time, and consequence.

7.4 Cabin pressure, through the work period

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · HOLD

SCENE 28W

The crew works in the middle cabin while an airlock cycles. A scrubber starts, and an old leak stays on the log. Track which change moves the pressure.

GUIDE 46W

Hold the cabin pressure inside the band on the console. The band narrows through the work period. The crew are moving toward the module with the suspect seal. A margin that was comfortable at the start is not comfortable there. The makeup valve is your control.

ASK 10W

Hold the cabin inside the band through the work period.

BEHIND THE BUTTON 125W

Why a leak is not a step. A leak is a rate: the cabin loses gas for as long as the hole is open, so the pressure keeps falling and does not settle anywhere. That is why a cabin with a small leak is held rather than corrected.

Why the airlock and the scrubber are different in kind. Cycling the airlock takes a volume of gas out once. The scrubber changes what the cabin is doing continuously. Both show up on the same gauge and want different answers.

Why the margin matters more later. Everything the crew are doing is reversible except being in the wrong module when the pressure goes. The band narrows because the cost of the same excursion rises as they move.

TAKES AS READ

a sealed volume loses pressure if gas escapes from it

VERDICT 118W

A leak is a rate, and everything about handling one follows from that. The cabin loses gas for as long as the hole is open. So the pressure keeps falling and never settles. A valve opened once and closed again buys a few minutes and changes nothing. What works is setting the makeup flow to match the loss and leaving it there. That is also what makes the leak measurable: the flow you need to hold steady is the leak rate. The ground needs that number to decide whether the module can stay in use. The airlock is the opposite case. Cycling it removes a fixed volume once, so it wants a correction rather than a new setting.

TAKEAWAY 17W

A leak is a rate, so it is answered by a setting rather than by a correction.

END OF THE SHIFT 17W

Use heat capacity, conduction, convection, radiation, energy balance as an evidence chain rather than as isolated facts.

SHIFT 8 OF 15

The Air Problem

BEFORE THE SHIFT — HUNT

Find the missing survival logs

Recover the air, water, power, fuel, heat, and filter logs so the room knows how long the crew can live.

PLAN CARD 42W

Shift 8. Carbon dioxide is rising even though the scrubber fan sounds normal. The spare filters do not fit, and the crew is starting to make small mistakes. You must find the blocked flow and force cabin air through a working filter.

BRIEFING 20W

Carbon dioxide is rising even though the scrubber fan runs. Find the blocked air path and build a safe fix.

WHAT YOU DECIDE 16W

Develop a safe temporary air-cleaning and circulation strategy using only validated physical principles and monitored limits.

8.1 Diagnose the air system

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CHAIN

SCENE 26W

Carbon dioxide keeps rising. The fan sounds normal and draws normal power, but little air crosses the filter. Use the pressure readings to find the block.

GUIDE 56W

Build the path the air takes, in order, and then name the link where it becomes limiting. The fan sounds normal and draws its usual current, which tells you the fan is working. Pressure on both sides of the filter and the bed is what locates the restriction, so read those rather than trusting the sound.

ASK 7W

Where does the air-transfer chain become limiting?

BEHIND THE BUTTON 147W

Why a working fan and low flow means a restriction. A fan draws current according to how much air it is actually moving. A fan meeting a blockage moves less air, works less hard, and sounds entirely normal. So the current reading is evidence that the fan is fine, not that the system is.

What a pressure difference across a component tells you. Flow through a restriction costs pressure. A large drop across the filter with a small one across the bed says the filter is the restriction, and the reverse says the bed is. That is the measurement that separates two very different repairs.

Why carbon dioxide is the clock. It rises whenever removal is slower than production, and the crew's symptoms begin well before the level is dangerous. The chain has to be diagnosed while everyone can still think clearly, which is the real deadline.

TAKES AS READ

a running motor does not guarantee useful flow · air must pass through every required link in a closed loop

VERDICT 81W

The fan supplies the pressure difference that can move air, but every downstream restriction decides how much flow that pressure can produce. Normal motor current therefore proves only that the fan is powered. It does not prove that cabin air reaches the sorbent. A large pressure drop across one link, paired with low total airflow, identifies the bottleneck. Because every carbon dioxide molecule must cross that link before reaching the scrubber, the whole loop performs at the rate that restriction allows.

TAKEAWAY 17W

A transport chain is limited by the required link that most restricts the quantity moving through it.

8.2 CO2 production scale

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · BALLPARK

SCENE 22W

Each crew member makes about 20 litres of carbon dioxide per hour. Find the six-hour total before sizing the new filter path.

GUIDE 62W

Five numbers, and two of them belong to a different question: a whole day, and a figure by mass rather than by volume. Ask of each whether it is a rate, a count, or a duration. What comes out is the minimum a scrubber has to keep up with. Anything slower than that only slows the rise rather than holding it steady.

ASK 8W

Estimate the carbon dioxide produced in six hours.

BEHIND THE BUTTON 111W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before the tile is placed. It is also how you catch a quantity that belongs to a different part of the problem. That habit is what this format exists to build.

TAKES AS READ

a rate multiplied by a time is a total

VERDICT 75W

Source rate sets the minimum removal rate. Three people at roughly 20 L per hour each produce about 60 L per hour together, or about 360 L over six hours at the stated reference conditions. This is a scale estimate, not a prediction of cabin partial pressure: cabin volume, ventilation, sorbent performance, and gas conditions still matter. The practical test is whether removal can at least keep pace with production while measured CO2 stops rising.

TAKEAWAY 7W

Source rate sets the minimum removal requirement.

8.3 Restore breathable air

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · CHOICE

SCENE 37W

Carbon dioxide is at 13 millimetres of mercury and still rising, and the limit is 15. The spare canister is square, but its socket is round. A sealed path has to be built from what is aboard.

GUIDE 60W

All four options describe something the fix could be judged on. Ask of each whether it changes how much air actually crosses the sorbent. Carbon dioxide is only removed from air that goes through, not past. A big canister with a gap around it does worse than a small one sealed tight. The fan makes the same noise either way.

ASK 16W

Tape, a suit hose, a sock and a flight-plan cover. What must the fix get right?

BEHIND THE BUTTON 201W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Here it ran to 88 per cent of passage quizzes, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason looks exactly like a sound one. It stops looking like one on the day the approximation is what is being tested.

TAKES AS READ

air, like water, takes the easiest path it is offered

VERDICT 78W

The fix needs both enough sorbent and a sealed flow path through it. Air that slips around the canister reaches the cabin unchanged. That bypass can look deceptively normal because the fan still runs and makes noise. Carbon dioxide is removed only from air that actually crosses the sorbent. A large canister with a gap around it can therefore perform worse than a smaller one with a tight seal. Flow path and chemical capacity have to work together.

TAKEAWAY 11W

A life-support fix must manage both chemical uptake and fluid flow.

END OF THE SHIFT 19W

Use ideal gases, partial pressure, flow resistance, pressure difference, filtration as an evidence chain rather than as isolated facts.

Communication Fades

PLAN CARD 38W

Shift 9. The radio loses 12 decibels when the ship turns. Ito, the communications lead, must tell a bad sender from a bad aim. You must restore the link before it is needed to guide the next burn.

BRIEFING 18W

The radio fades when the ship turns. Find the cause, restore the link, and save the tracking signal.

WHAT YOU DECIDE 14W

Restore a reliable low-bandwidth link and use the signal itself as a navigation measurement.

9.1 Why did the signal fade?

WAVES & COMMUNICATIONS · CAPSULE COMMUNICATIONS · A A PLACE · CONTROL

SCENE 42W

Voice and telemetry drop about twelve decibels at both ground stations whenever the ship turns. The spacecraft transmitter output is nominal. Three things can be tested without committing the mission: the antenna pointing, the redundant transmitter chain, and a backup receiver path.

GUIDE 70W

The number you are watching is the received signal strength, currently about twelve decibels low at both stations. Each test swaps or corrects one part of the path without committing the mission. A response counts only if it is larger than the noise on the link. Two stations failing together already narrows what can be common to both. Run one test, undo it, and name the part the fade follows.

ASK 14W

Which reversible change makes the twelve-decibel drop in received carrier power appear and disappear?

BEHIND THE BUTTON 178W

Why two stations at once is the clue. The two ground stations are on opposite sides of the Earth, with separate receivers, separate crews and separate weather. A fault in one of those cannot show up at the other. So anything that explains both has to be common to both. That means the spacecraft end, or the geometry between them.

What twelve decibels means. Decibels are a ratio on a logarithmic scale. Three of them is about half the power. Ten is a tenth, and twelve is about one sixteenth of the power arriving. That is the difference between voice you can work with and voice you cannot, which is why the number matters more than it sounds.

Why every test is reversible here. There is a crew aboard, so a test that cannot be undone is not a test but a decision. The pointing correction, the redundant transmitter chain and the backup receiver path were chosen because each can be tried and put back. That is what makes the reversal half of the method available at all.

TAKES AS READ

decibels add gains and losses on a link budget · a causal test changes one condition while the others stay fixed · kinematics — taken as read

VERDICT 79W

Several things can weaken a radio link, but a causal test asks whether the signal follows one controlled change. Swapping to a backup ground receiver changes almost nothing, so the fade is not local to one receiver. Swapping transmitter chains at the same output also changes almost nothing. Correcting the antenna pointing restores nearly all the missing power. Reversing that pointing change brings the loss back. That two-way response ties the fade to antenna geometry rather than to coincidence.

TAKEAWAY 18W

A reversible one-variable test can separate a causal fault from conditions that merely occur at the same time.

9.2 Wavelength of the radio link

WAVES & COMMUNICATIONS · CAPSULE COMMUNICATIONS · A PERSON · BALLPARK

SCENE 31W

The link is fading, and the pointing model needs the carrier wavelength. The ship transmits at 2.0 gigahertz. Both ground stations are watching the received power, so the beam width matters.

GUIDE 61W

Five numbers, and three of them are traps. One is the carrier misread as megahertz. One is the speed of sound. One is the observed loss. Ask of each whether it belongs to this wave at all. And note what the answer feeds: the beam width scales with wavelength over aperture, so this number sets how forgiving the pointing can be.

ASK 3W

Estimate the wavelength.

BEHIND THE BUTTON 111W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before the tile is placed. It is also how you catch a quantity that belongs to a different part of the problem. That habit is what this format exists to build.

TAKES AS READ

frequency and wavelength are two descriptions of one wave · kinematics — taken as read

VERDICT 78W

Wavelength is wave speed divided by frequency. At 2.0 gigahertz, the radio wavelength is about 0.15 metre. That number also sets the scale of the antenna beam. For a fixed dish size, beam width is roughly proportional to wavelength divided by aperture. A longer wavelength gives a wider, more forgiving beam. A shorter wavelength gives a narrower beam and demands better pointing. The calculation is simple, but it feeds directly into the pointing tolerance used for the link.

TAKEAWAY 11W

Frequency and wavelength are reciprocal descriptions of the same propagating wave.

9.3 Recover the weak link

ROTATION & STRUCTURES · STRUCTURES TEST BAY · A A PLACE · SEQUENCE

SCENE 27W

The radio is weak and its pitch is moving. The crew can barely hear the ground. Put the checks in the order that can restore the link.

GUIDE 60W

All four steps will be taken, so ask which buys the most for the least. Pointing is the largest term in the link budget and costs nothing but attitude. Narrowing the bandwidth buys signal without touching the spacecraft. And one of these four is not a recovery action at all: it is what the recovered data are then used for.

ASK 14W

Put the four cards in order. Start with the step that must happen first.

BEHIND THE BUTTON 169W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards have nothing before or after them. That constraint usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a slower signal needs less power to be read · waves: frequency, wavelength, speed — taken as read

VERDICT 78W

Recovery combines three disciplines in a fixed order, cheapest and largest first. Pointing is the biggest term in the link budget and costs nothing but attitude, so geometry is checked before anything else. Reducing the data rate and narrowing the receiver bandwidth buys signal-to-noise without touching the spacecraft. Coordinating the large ground antennas with predicted Doppler puts the receiver where the signal actually is. And the range and Doppler recovered along the way feed straight back into navigation.

TAKEAWAY 8W

Communication recovery joins wave physics, control, and estimation.

9.4 What failed: the spacecraft or the sensors? — Review

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CALLBACK · TRACE

SCENE 35W

Three thermocouples on the fuel tank climb together, four hours after a signal-conditioning card was swapped. The boil-off rate and the crew's hand-held reading stay steady. Carter wants the diagnosis before anybody vents the tank.

GUIDE 57W

Open each temperature channel to see what it depends on. Keep the ones that would still stand if the new card were faulty, and untick the rest. Then name the shared source. Venting the tank is not free and it is not reversible, so the question is whether the gas is warming or the measurement chain is.

ASK 14W

Which temperature evidence survives the dependency check, and what shared source moved the rest?

BEHIND THE BUTTON 164W

Why three thermocouples can be one measurement. Redundant temperature channels are usually amplified and digitised by one conditioning card, which supplies the excitation and the cold-junction reference. A fault in either moves all three readings the same way at the same moment, which is exactly what a warming tank looks like on the display.

Why the boil-off rate is worth more than three thermocouples here. Gas leaving a tank through a relief path is a flow. It is measured by a different instrument on a different chain, and it is a direct consequence of the tank being warm. It cannot inherit a fault in the temperature electronics because it does not pass through them.

Why the timing matters as much as the pattern. The card was fitted this morning, and the divergence starts hours later rather than at the swap, which is how a marginal connection behaves as it warms. A shared cause does not have to act at the instant it is installed.

TAKES AS READ

redundant sensors often share one amplifier and one reference · an instrument on a different physical principle does not inherit that fault

VERDICT 97W

Three channels agreeing is three votes only if they can fail separately, and these three are amplified and referenced by one card. A fault in that card moves all of them the same way at once, which is indistinguishable on the display from a tank that is genuinely warming. The boil-off rate is a flow on a different instrument, and the crew's hand-held reading is an optical measurement made in the cabin. Neither passes through the card, and both say the gas is where it was. Venting would have thrown away propellant to correct an electrical fault.

TAKEAWAY 18W

> Redundancy only buys independence when the redundant paths do not meet, and they usually meet at the card.

END OF THE SHIFT 17W

Use waves, frequency, wavelength, inverse-square behavior, Doppler effect as an evidence chain rather than as isolated facts.

SHIFT 10 OF 15

A Blind Maneuver

PLAN CARD 49W

Shift 10. The main guidance unit is still off, and bright debris hides many stars. The crew will aim through a window with Earth and the Sun as guides. You must tell a real turn from a sighting shift before a tiny angle becomes a large miss at Earth.

BRIEFING 19W

The crew must aim a burn without the main guidance unit. Bound the sighting errors before trusting the turn.

WHAT YOU DECIDE 11W

Design a manual alignment procedure whose dominant angular errors are bounded.

10.1 What shifts the apparent direction?

WAVES & COMMUNICATIONS · CAPSULE COMMUNICATIONS · A A PLACE · PROTOCOL

SCENE 29W

Bright debris hides many stars. The crew will use Earth, the Sun, and a window grid. First tell a true ship turn from a shift caused by eye position.

GUIDE 42W

Four apparent shifts, four causes. Ask what physically moved: the eye, the optics, the viewpoint, or the spacecraft. The crew is about to aim a burn with this sight, so an apparent reticle shift cannot be allowed to masquerade as attitude error.

ASK 24W

Match each apparent shift to what physically moved, and keep angular error apart from a real turn of the ship. Use each choice once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation. Choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced. The fault is usually not in the join you are struggling with. It is in one you made early and stopped questioning.

TAKES AS READ

what you see through an eyepiece depends on where your eye is

VERDICT 83W

Manual alignment works only if instrument geometry is separated from vehicle motion. Moving the eye relative to a nearby reticle creates sighting parallax without changing the external line of sight. Focus changes sharpness, not direction. A nearby object that changes apparent position against a distant reference as viewpoint changes shows geometric parallax. A stable celestial reference moving across the reticle instead indicates a change in body orientation. These effects can look similar in one glance and have completely different consequences for a burn.

TAKEAWAY 9W

Manual navigation requires separating instrument geometry from spacecraft motion.

10.2 Reading the trace instead of the number

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · PROTOCOL

SCENE 28W

The board shows 90 minutes of range data. The latest point is only one clue. Use the line's shape to find speed, force, noise, or a bad offset.

GUIDE 65W

Four features of one plotted line, and four things they could mean. Pair them by asking what changes and what does not. A slope is a rate. A change of slope means something acted. Scatter around a smooth line is the instrument. A step with the same slope after it is bookkeeping, and that is the one the room will explain with an invented manoeuvre.

ASK 15W

Match each feature of the plotted trace to what it tells you about the motion.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation. Choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

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TAKES AS READ

the slope of a distance-time graph is a speed · kinematics — taken as read

VERDICT 83W

A distance-time graph contains more information than its newest point. The slope is the rate of change, so it gives speed. A change in slope is acceleration and signals that a force acted. Random scatter around a smooth trend is measurement noise rather than real motion. A sudden step followed by the same slope is different. It suggests an offset in calibration, units, or bookkeeping. Reading the shape of the trace prevents the room from inventing a maneuver to explain an instrument change.

TAKEAWAY 17W

A graph carries rate and change of rate, neither of which is visible in the latest reading.

10.3 How far it drifts while nobody is steering

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · BALLPARK

SCENE 23W

A small push of 0.004 m/s^2 acts for 90 seconds. Find how far the ship drifts before the team decides if it matters.

GUIDE 67W

Four numbers, and one of them is surface gravity, offered for scale rather than for use. Another is the one-half that belongs in the relationship. Ask of each whether it is an acceleration, a time, or a constant. And note the square on the time. Double the coast and the drift is four times larger, which is why the coast length is the thing to argue about.

ASK 7W

Estimate the displacement over the ninety-second coast.

BEHIND THE BUTTON 111W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before the tile is placed. It is also how you catch a quantity that belongs to a different part of the problem. That habit is what this format exists to build.

TAKES AS READ

an object under constant acceleration covers half the acceleration times the time squared · vectors and components — taken as read

VERDICT 70W

For constant acceleration from rest, displacement grows as one-half the acceleration times time squared. Here the result is about 16 metres. The square on time is what makes the effect important. Double the coast time and the drift becomes four times larger. A six-minute coast therefore produces far more than twice the displacement. The same mathematics describes free fall and projectile motion whenever one approximately constant acceleration dominates the motion.

TAKEAWAY 12W

Under a constant acceleration, displacement grows with the square of the time.

10.4 What the pressure difference is for — Review

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · CALLBACK · CHOICE

SCENE 21W

Coolant flow falls by one third, but the pump still adds 24 psi. Decide whether the push or the path changed.

GUIDE 60W

All four options explain a flow that has fallen. They differ in what they hold responsible: the path, the pump, the instrument, or the fluid. Ask of each whether it is consistent with a driving difference that has not moved. A pressure difference makes the fluid go; how obstructed the path is decides how much of it goes per minute.

ASK 22W

The pump still holds 24 psi across the loop and the flow has fallen by a third. What does that combination mean?

BEHIND THE BUTTON 170W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why an unchanged difference is the informative half. Two numbers describe this loop: what pushes and what resists. If the push is where it was and the flow has fallen, the resistance is the only term left to have moved. Had the difference fallen with the flow, the pump would be the first thing to look at instead.

What can add resistance in a cold loop. A partly closed valve, a loaded filter, gas collected at a high point, or glycol thickening as the cabin cools. All four reduce the flow at an unchanged pressure difference. What separates them is where in the loop the pressure is lost, rather than how much.

TAKES AS READ

a pressure difference drives a flow and the path decides its rate · newton's laws and free-body reasoning — taken as read

VERDICT 89W

Two quantities describe this loop and only one of them has moved. The pressure difference is what drives the glycol, and the resistance of the path decides how much arrives per minute. With the difference held at twenty-four psi and the flow down by a third, the resistance is the only term left that can have changed. A part-closed valve, a loaded filter, gas trapped at a high point or glycol thickening in a cooling cabin all do that. A healthy pump guarantees a pressure rise, never a flow.

TAKEAWAY 15W

When the push has not changed and the flow has, the path is what moved.

END OF THE SHIFT 18W

Use lenses, angular measurement, parallax, reference frames, error propagation as an evidence chain rather than as isolated facts.

SHIFT 11 OF 15

Crossing the Atmosphere

PLAN CARD 43W

Shift 11. The capsule will strike the air near 11 kilometres per second. Too shallow may send it back toward space, while too steep may crush or burn the crew. You must turn a range of possible paths into one safe entry call.

BRIEFING 21W

The ship is nearing a narrow entry path. Choose a target that stays safe when the air and tracking are uncertain.

WHAT YOU DECIDE 12W

Choose a corridor and monitoring strategy robust to atmospheric and navigation uncertainty.

11.1 Where orbital energy goes

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · SEQUENCE

SCENE 23W

The capsule reaches the air with huge kinetic energy. It must lose nearly all of it before splashdown. Trace where that energy goes.

GUIDE 64W

These four are one chain, and reentry is an energy problem before it is a temperature problem. Ask of each card what has to be true first. There is no atmospheric braking before there is atmosphere. Energy cannot become heat until it has left the vehicle's motion. The last card is where the shield and the trajectory decide the rates rather than the total.

ASK 14W

Put the four cards in order. Start with the step that must happen first.

BEHIND THE BUTTON 169W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards have nothing before or after them. That constraint usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

energy is not destroyed, it is moved somewhere else · newton's laws and free-body reasoning — taken as read

VERDICT 75W

Reentry is primarily an energy-transfer problem. At atmospheric interface the vehicle's mechanical energy is dominated by its enormous kinetic energy. Aerodynamic forces do negative work on the spacecraft, transferring mechanical energy into internal energy of the shocked gas, the wake, the heat shield, and the structure. The heat shield does not make that energy disappear. Its shape and materials control where the heating occurs and how much reaches the cabin. The trajectory controls the rate.

TAKEAWAY 11W

Reentry is an energy-dissipation problem constrained by human and material limits.

11.2 Kinetic-energy scale

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A PERSON · BALLPARK

SCENE 20W

A 5,000 kg capsule enters at about 11,000 m/s. Find the energy that drag and the heat shield must handle.

GUIDE 60W

Five numbers, and two of them belong to other entries: the speed from low orbit, and surface gravity. Ask of each whether this energy depends on it. And note the square on the speed. Forty per cent more speed is nearly twice the energy at the same mass. That is the whole difference between the two entries on the list.

ASK 7W

Estimate the kinetic energy to be dissipated.

BEHIND THE BUTTON 111W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before the tile is placed. It is also how you catch a quantity that belongs to a different part of the problem. That habit is what this format exists to build.

TAKES AS READ

kinetic energy depends on mass and on speed · newton's laws and free-body reasoning — taken as read

VERDICT 77W

Kinetic energy is one-half the mass times speed squared. The squared speed is the dominant feature of lunar return. Raising entry speed by about 40% nearly doubles the kinetic energy for the same mass. That extra energy must still be removed before landing. The heat shield therefore faces a much harder job even if the capsule itself is unchanged. Comparing speeds without squaring them would badly understate the difference between low-Earth-orbit entry and return from the Moon.

TAKEAWAY 8W

Speed dominates kinetic energy because it enters squared.

11.3 Protect the entry corridor

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CLOUD

SCENE 26W

Safe entry lies from 5.3° to 7.7°. The best path is 6.2° with a 0.5° one-sigma spread. Use two actions to make the full range safer.

GUIDE 58W

The display is a probability distribution, not a hard band of every possible angle. Its centre is 6.2 degrees and its one-sigma width is 0.5 degrees; the safe corridor spans 5.3 to 7.7. One action shifts the centre, while an independent tracking pass reduces the statistical width. Apply the actions and judge the probability left outside either boundary.

ASK 27W

Can you make the entry solution robust without confusing a better center with a narrower uncertainty? Place the bars to report the mean and its one-sigma uncertainty.

BEHIND THE BUTTON 161W

Why one sigma is not a fence. A standard deviation describes the width of a probability distribution; it is not the set of all angles the vehicle could have. Entry safety therefore depends on how much probability remains outside the corridor, not simply whether the drawn one-sigma interval fits between the lines.

Two different actions. A targeting correction moves the nominal, which slides the whole band along the corridor. A tracking pass adds information, which is the only thing that makes it narrower. Confusing them is the classic error in this room: the number on the display gets better and the risk does not.

Why 'when is it ready' is part of the question. Every action costs time on a clock the vehicle sets, and the crew cannot wait for an arbitrarily tight solution. So the decision is not merely what to do but when the band is inside the corridor by enough that the next hour is better spent elsewhere.

TAKES AS READ

an uncertainty band describes a range of plausible states · a narrower distribution puts more probability near its center

VERDICT 79W

The nominal entry angle is only the centre of a distribution. A targeting correction moves that centre but does not by itself make the state better known. Independent tracking can narrow the distribution by adding information. Here the authored Gaussian model requires both. Recentering removes the offset. Tracking reduces the width, until the probability inside the 5.3–7.7 degree corridor exceeds the stated 99.5% decision rule. The rule is a model assumption for this game, not a universal reentry standard.

TAKEAWAY 17W

> Moving an estimate changes its center; new information changes its width, and safety can depend on both.

END OF THE SHIFT 16W

Use kinetic energy, drag, work-energy, heating, deceleration as an evidence chain rather than as isolated facts.

SHIFT 12 OF 15

The Structure Is Vibrating

PLAN CARD 48W

Shift 12. A pump makes one panel shake hard near 3,200 rpm. A line behind the panel has rubbed for an hour, and the motion grows each time the pump returns to that speed. You must test for resonance and move away from it before the line breaks.

BRIEFING 20W

A pump is making a panel shake near one speed. Move the system away from resonance before a line breaks.

WHAT YOU DECIDE 15W

Move the system away from resonance and verify that the mitigation works across operating conditions.

12.1 What is driving the vibration?

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CHOICE

SCENE 27W

The panel shakes hard only near 3,200 rpm. The motion fades above and below that speed. Okoye, the structures engineer, asks if the pump is driving resonance.

GUIDE 51W

Three facts have to agree: the response peaks near 3,200 rpm, it falls on either side, and two independent accelerometers see the same mode. Ask which explanation predicts all three. A loose sensor cannot make a second sensor respond, and an impact would not switch on and off with rotor speed.

ASK 12W

Which explanation fits the frequency, the speed dependence and the independent accelerometers?

BEHIND THE BUTTON 201W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Here it ran to 88 per cent of passage quizzes, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason looks exactly like a sound one. It stops looking like one on the day the approximation is what is being tested.

TAKES AS READ

every structure has speeds at which it prefers to shake · centre of mass and balance — taken as read

VERDICT 75W

Resonance is identified by the relationship between forcing frequency and structural response. A rotor at 3,200 rpm forces the structure at about 53.3 Hz. If a structural mode lies near that frequency, repeated forcing can build a large response even when the static load is harmless. Moving the pump away from the band immediately reduces the forcing at that mode. The second accelerometer confirms the response belongs to the structure rather than one loose sensor.

TAKEAWAY 14W

How hard something shakes matters less than what is shaking it and how fast.

12.2 Does the natural frequency match the pump?

ROTATION & STRUCTURES · STRUCTURES TEST BAY · A A PLACE · BALLPARK

SCENE 27W

Treat the panel as a 4 kg effective mass on an equivalent stiffness of 4.5×10^5 N/m. Find its natural period, and compare the frequency with the pump.

GUIDE 40W

The component's mode depends on effective mass and stiffness. First calculate the period from $2\pi\sqrt{(m/k)}$, then invert it for frequency and compare with 3,200 rpm ÷ 60. A resonance claim only earns its place if the two frequencies actually meet.

ASK 15W

Estimate the natural period, then compare the natural frequency it implies with the pump forcing.

BEHIND THE BUTTON 111W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before the tile is placed. It is also how you catch a quantity that belongs to a different part of the problem. That habit is what this format exists to build.

TAKES AS READ

a stiffer spring vibrates faster and a heavier mass vibrates slower · newton's laws and free-body reasoning — taken as read

VERDICT 59W

For the simplified spring-mass mode, $T = 2\pi\sqrt{(m/k)}$. With 4 kg and 4.5×10^5 N/m, the period is about 0.0187 s, or a natural frequency of about 53.4 Hz. The pump at 3,200 rpm forces at about 53.3 cycles per second. Those numbers now match within the precision of the model, which is exactly the evidence the resonance diagnosis needs.

TAKEAWAY 17W

A resonance claim requires the forcing frequency and natural frequency to match, not merely a large vibration.

12.3 Stop the resonance

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · CHOICE

SCENE 27W

The panel's motion grows each time the pump reaches 3,200 rpm. The pump must stay on, but its speed can change now. Choose the safest first move.

GUIDE 60W

Resonance needs two things: a structure, and a forcing frequency. All four options act on one of them, or on neither. Ask of each which, and how fast it can be done from where the crew is sitting. The amplitude has already grown for three cycles. So one option's assumption, that the structure survives the passage, is no longer safe.

ASK 7W

What is the safest immediate control-panel action?

BEHIND THE BUTTON 201W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Here it ran to 88 per cent of passage quizzes, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason looks exactly like a sound one. It stops looking like one on the day the approximation is what is being tested.

TAKES AS READ

a resonance needs both a structure and something driving it · newton's laws and free-body reasoning — taken as read

VERDICT 73W

Resonance requires periodic forcing near a natural frequency. The fastest reversible control is therefore to move the pump rotor away from the measured resonant band while preserving the required flow if the pump map allows it. Added damping or structural changes could be permanent fixes, but they require hardware access. Racing through the band assumes the panel and rubbing line survive the transient; the growing amplitude gives no reason to make that assumption.

TAKEAWAY 16W

A structural fix should change the dynamics and then demonstrate that the dangerous mode is controlled.

12.4 What the room is calling now

WAVES & COMMUNICATIONS · CAPSULE COMMUNICATIONS · A A PLACE · SPOT

SCENE 28W

Calls reach the flight loop faster than you can answer. The top risk changes after the burn, the leak, and crew wake-up. Follow the rule now in force.

GUIDE 58W

Answer the calls the current rule wants and hold the rest. The rule is called once on the loop and then it stands; nobody repeats it. What is scored is the calls either side of a change. They are the only ones that show whether the room is working to this rule or to the one before it.

ASK 24W

Answer to the rule in force. It is set by the failure mode that matters most right now, and it changes without being repeated.

BEHIND THE BUTTON 91W

Why a flight rule changes. Before a burn everything defers to the burn. Once a leak is confirmed everything defers to the leak. When the crew are asleep the rule is different again, because a call that wakes them costs something a call to the ground does not.

Why the changeover is the dangerous part. A room working to the old rule for two more minutes gives the loop to a call that no longer matters. The call that does matter waits behind it, with nothing to say it was displaced.

TAKES AS READ

a control room answers calls in an order somebody decides

VERDICT 115W

Three rules run across the shift, and each wants a different part of the board. Anything affecting the burn, then anything affecting the leak, then anything that would wake the crew. A call can answer two at once, which is what makes the change cost real rather than notional. The panel scores the window either side of each change. Most of what arrives is wanted by neither rule, and is correctly held by somebody who has read nothing. The room's own version is the two minutes after a leak is confirmed. The rule has changed, and the loop is still going to the trajectory people, because that is who it belonged to a moment ago.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

END OF THE SHIFT 19W

Use simple harmonic motion, natural frequency, resonance, damping, forced oscillation as an evidence chain rather than as isolated facts.

SHIFT 13 OF 15

Choose the Return Path

BEFORE THE SHIFT — CANVASS

Find who truly saw the explosion

Ask each console what it measured so one bad data path is not mistaken for damage across the ship.

PLAN CARD 38W

Shift 13. Four paths home remain, and each spends a different safety margin. Water may run out almost a day earlier than the best estimate. You must choose the path that still works when that estimate is wrong.

BRIEFING 21W

Four paths home spend fuel, time, water, and entry margin in different ways. Choose the one that survives a bad estimate.

WHAT YOU DECIDE 13W

Select a return plan with explicit margins and contingencies for the dominant uncertainties.

13.1 Name the binding constraint

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · PROTOCOL

SCENE 24W

Four paths home are on the board. Each spends fuel, time, water, heat, or radio cover. Name the limit each path comes closest to.

GUIDE 54W

Four return paths, and four subsystems that pay for them. Pair them by asking what each path spends: consumables, propellant, heating, or the crew's own ability to navigate. No path is cheapest on every count. So the useful question is not which is fastest but which constraint each one pushes closest to its limit.

ASK 12W

Match each case to the best scientific answer. Use each choice once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation. Choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced. The fault is usually not in the join you are struggling with. It is in one you made early and stopped questioning.

TAKES AS READ

a spacecraft carries a fixed amount of everything it needs

VERDICT 77W

A trajectory is a system choice because its costs land on different subsystems. More flight time spends oxygen, water, scrubber capacity, and electrical energy. A larger correction burn spends propellant and increases execution error. A steeper atmospheric entry raises peak heating and deceleration. A long communications blackout leaves the crew without ground support when navigation is most difficult. No single path minimizes every cost. The useful question is which constraint each path pushes closest to its limit.

TAKEAWAY 12W

The best physical solution is not necessarily the shortest or lowest-fuel solution.

13.2 Select the robust trajectory

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · STRESS

SCENE 32W

The four paths take 58, 63, 66, or 71 hours, and their propellant and entry margins differ. The consumable endurance is uncertain by about a day, and water is the tightest supply.

GUIDE 58W

Four return paths, and the consumable estimate is uncertain by about a day. Move that estimate to its pessimistic end before choosing, and watch which trajectories go dark. A path that works at the nominal estimate and fails a day short is not a plan. Water is the tightest supply, so it is the one to test against.

ASK 14W

Which trajectory still works when the life-support estimate is wrong in the bad direction?

BEHIND THE BUTTON 142W

Why the pessimistic end rather than the expected one. The endurance figure is an estimate with a spread, not a measurement. Choosing at the middle of it means accepting a real chance of running out. Running out of water on a return trajectory is not recoverable by any later decision.

What the trade actually is. A faster path uses more propellant and leaves less entry margin; a slower one asks more of the consumables. Neither end is safe, which is why the choice is made by testing both against their uncertain quantity rather than by minimising one number.

Why water rather than oxygen or power. It is the supply with the least margin, so it fails first, and every hour added to the trajectory costs it directly. The binding constraint is the one worth stressing; testing against a comfortable supply proves nothing.

TAKES AS READ

a constraint can eliminate an otherwise attractive option · uncertainty should be tested in the direction that makes the plan hardest

VERDICT 84W

A nominal comparison hides which path fails first when an uncertain input moves. The shortest return looks best if time is the only score, but its entry margin is too small. The longest options preserve other resources but depend on life support lasting beyond the pessimistic estimate. The balanced path is not the nominal winner on speed. It is the only candidate that keeps enough entry and propellant margin while also returning before the low-end consumable limit. That is robustness rather than average performance.

TAKEAWAY 20W

A robust choice survives the plausible error in the assumption that matters most, even when another option looks better nominally.

13.3 Commit to the path

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · SEQUENCE

SCENE 26W

The burn will close most other paths. Before it starts, write the sign that would prove this plan unsafe while there is still time to act.

GUIDE 55W

All four of these will happen, and the burn closes most of the alternatives. Ask of each what would be impossible afterwards. Comparing candidates under the nominal case settles nothing, because the nominal case was never in doubt. And an abort trigger defined after the burn is a trigger with nothing left to act on.

ASK 13W

Order the commitment so the abort trigger still has something to act on.

BEHIND THE BUTTON 169W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards have nothing before or after them. That constraint usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a decision can only be undone while the means to undo it still exist

VERDICT 89W

A plan is a hypothesis about future physical states, and it needs a way to be found wrong in time. The margins are quantified first, because they are what the decision is made of. The candidates are compared under off-nominal conditions rather than nominal ones, because the nominal case is the one that was never in doubt. The abort trigger is named before the burn — defined afterwards it is a trigger with nothing left to act on. Then the margins keep being updated against what the tracking returns.

TAKEAWAY 16W

An abort trigger has to be defined while there is still propellant to act on it.

END OF THE SHIFT 17W

Use optimization, constraints, trade spaces, margins, critical path as an evidence chain rather than as isolated facts.

SHIFT 14 OF 15

The Last Correction

PLAN CARD 39W

Shift 14. Several ground results now place the ship ahead of its path. They agree, but they also use the same clock. You must decide whether the ship moved or the clock moved before the last burn window closes.

BRIEFING 21W

Ground tracking says the ship moved, but the stations share one clock. Decide whether to burn or check the clock first.

WHAT YOU DECIDE 14W

Decide whether to burn using the expected benefit relative to navigation and propulsion uncertainty.

14.1 Real trajectory error or common clock drift?

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CHOICE

SCENE 27W

Range and Doppler place the ship ahead of its path. Both use one ground clock. Compare them with the star angle and the clock log before burning.

GUIDE 34W

Several radiometric products agree, but agreement is not independence. Ask which products inherit the same station time-and-frequency reference, then compare them with the optical angle and an independent timing standard before authorising a burn.

ASK 16W

Which explanation fits the range, angle and Doppler evidence, and the independent timing check beside it?

BEHIND THE BUTTON 201W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading. A step skipped. A quantity confused with a rate. A correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance. Here it ran to 88 per cent of passage quizzes, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well. A right choice made for an approximate reason looks exactly like a sound one. It stops looking like one on the day the approximation is what is being tested.

TAKES AS READ

several instruments can be wrong together if they share a part

VERDICT 102W

Agreement matters only when measurements can fail independently. The range and coherent-Doppler products are reduced against the same ground time-and-frequency reference. A timebase error can therefore bias the derived state in a correlated way, even though the raw observables are different. The optical angle uses different physics and still matches prediction. An independent timing check then shows the common reference is offset, and reprocessing the radiometric data removes most of the apparent trajectory jump. The lesson is not that every clock error moves range and Doppler identically. It is that derived products can share a reference, and therefore share a failure mode.

TAKEAWAY 18W

When several measurements agree, ask whether they agree independently or merely inherit the same clock, calibration, or model.

14.2 Combine the orthogonal error components

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A PERSON · BALLPARK

SCENE 26W

The navigation covariance gives one-sigma position errors of 6 km and 8 km on perpendicular axes. Find the root-sum-square scale, without calling it a 68% circle.

GUIDE 39W

The two components are independent and perpendicular, so their variances add. The square root of 6^2+8^2 is a useful RMS/RSS size for the two-dimensional error vector. It is not, by itself, the radius containing 68% of a two-dimensional Gaussian.

ASK 5W

Estimate the root-sum-square position-error scale.

BEHIND THE BUTTON 111W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. Two things are being tested. Can you pick the quantities the relationship actually needs? And does the size of the answer look right once they are in place?

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch is caught before the tile is placed. It is also how you catch a quantity that belongs to a different part of the problem. That habit is what this format exists to build.

TAKES AS READ

the two quoted axis errors are independent one-sigma components

VERDICT 69W

Independent one-sigma components along perpendicular axes combine in quadrature as an RMS scale: $\sqrt{(6^2+8^2)}=10$ km. That preserves the variance geometry and is useful for comparing error budgets. But a two-dimensional Gaussian does not put 68% of its probability inside a circle of that radius. A true radial confidence region comes from the full covariance. The game therefore labels 10 km as an error scale, not a one-sigma radial boundary.

TAKEAWAY 20W

Independent orthogonal variance components add in quadrature, but their RSS is not the same thing as a radial confidence interval.

14.3 Burn or observe?

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · VALUE

SCENE 27W

Two tracking results use the same clock. A burn is ready, but one more check can still fit. Buy evidence that can fail in a different way.

GUIDE 56W

Two tracking solutions disagree, and both are computed through the same station clock model. Open each piece of evidence and ask what its result would change about the burn decision. Buy the evidence that could distinguish a real trajectory error from a clock error, not the evidence that would refine a number both solutions already share.

ASK 10W

Which evidence is worth buying before committing the correction burn?

BEHIND THE BUTTON 134W

Why two disagreeing solutions can share a fault. Both are derived from timing measurements referred to the same clock model. A modelling error moves both. It can move them by different amounts, because they use different station geometries. That looks exactly like a genuine disagreement about where the spacecraft is.

What would separate the two explanations. A measurement that does not depend on that clock model: an optical sighting, a different tracking technique, a station whose timing comes from elsewhere. Any of those either confirms the trajectory or exonerates it.

Why the burn makes it urgent. Correcting a trajectory that is not wrong introduces the error you were trying to remove, and spends propellant doing it. The decision closes shortly, which is what makes the choice of evidence a decision rather than a preference.

TAKES AS READ

two measurements that share a clock do not provide independent timing evidence · an observation can be valuable even if it is less precise · newton's laws and free-body reasoning — taken as read

VERDICT 78W

More precision on the same dependency does not resolve a common-mode ambiguity. Reprocessing the same timing data can reduce numerical noise while leaving the clock error untouched. Another range pass tied to that clock has the same weakness. An optical star-angle observation uses different physics and a different reference, so it can tell whether the spacecraft itself is displaced. Under a small budget, that independence is worth more than another precise copy of the evidence already in hand.

TAKEAWAY 17W

The most valuable measurement is the one whose failure mode is independent enough to change the decision.

END OF THE SHIFT 23W

Use random and systematic error, weighted averages, errors that move together, propagation, decision thresholds as an evidence chain rather than as isolated facts.

SHIFT 15 OF 15

Reentry

PLAN CARD 50W

Shift 15. Entry starts in 11 minutes, and the ground will soon lose the radio for four minutes. The shield, flight path, guidance unit, battery link, and hand aim must all work together. You have two checks left, then you must send the crew into the air or call no-go.

BRIEFING 20W

Entry starts in minutes and the radio will soon go silent. Spend the last checks and make the final call.

WHAT YOU DECIDE 16W

Spend the last verification checks, lock the blackout sequence, and make the final go/no-go entry decision.

15.1 Disposition final readiness

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · ATTEST

SCENE 28W

Entry starts in 11 minutes. Four claims are signed, but one lacks a check outside its own team. Spend the last two checks where failure would hurt most.

GUIDE 55W

Four signed readiness claims, eleven minutes, and two physical checks that can still be completed. Open each claim and read what backs it. Some rest on independent evidence and one does not. Hold what the evidence does not support, and spend the checks where being wrong would matter most once the vehicle is in blackout.

ASK 12W

Which claims deserve the final physical check before the vehicle enters blackout?

BEHIND THE BUTTON 137W

Why blackout changes the arithmetic. During entry there is no telemetry and no way to intervene. Every claim on that console is a decision that cannot be revisited for several minutes. That is why an unverified claim is worse here than anywhere else in the mission.

What a signature means at this point. Somebody has taken responsibility for a statement. Under time pressure, signatures are given on the basis of what somebody was told. That is how a claim with nothing behind it takes on the same appearance as one with a test behind it.

How to choose between two unbacked claims. By consequence during entry, not by how doubtful each is. A claim whose failure is survivable is worth less verification than one that is not, however much more likely the first is to be wrong.

TAKES AS READ

a signature records responsibility, not physical truth · verification time should be spent according to consequence and existing evidence

VERDICT 60W

A signature records responsibility, not physical truth. Verification time should go where consequence is highest and independent evidence is weakest. The undocumented shield-state change is therefore the first check. The entry state is already well supported, but its consequence is so large that it earns the second final check. The guidance self-test and corrected station bias already have direct evidence.

TAKEAWAY 21W

A signed readiness claim is a record of what someone believes; critical physical states still need evidence proportional to their consequence.

15.2 Execute the final physical chain

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · SEQUENCE

SCENE 19W

Entry starts in 11 minutes. The radio will go dark for four. Put each ground-led step before that silence.

GUIDE 63W

Four minutes with no radio is what orders these four. Ask of each whether it needs ground input, and whether it can be done after the interface. Anything the ground has to approve happens before the blackout. Anything compared against a prediction can only happen once entry has begun. And the record is the only direct test the predicted envelope will ever get.

ASK 10W

Order the entry around the four minutes with no radio.

BEHIND THE BUTTON 169W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards have nothing before or after them. That constraint usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

you cannot fix from the ground what happens while the radio is quiet

VERDICT 79W

The blackout forces the order of operations. Attitude, configuration, and the approved state estimate are set before atmospheric interface because ground commands will soon be unavailable. The roles of onboard guidance and the ground are also settled before commitment. Deceleration and heating can only be compared with predictions after entry begins. Recovery follows once the high-energy phase is over. The telemetry record is preserved throughout, because this single flight is the only direct test of the predicted entry envelope.

TAKEAWAY 17W

Entry is ordered by the blackout: whatever needs the ground has to be settled before it starts.

15.3 Give the entry GO

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · CHOICE

SCENE 28W

The shield matches its safe setup. Two kinds of tracking keep the ship inside the entry path. Guidance can fly through the radio blackout. Make the final call.

GUIDE 52W

This is the decision the previous fourteen shifts built. Ask whether any unresolved item is both capable of changing the entry outcome and still actionable before interface. Do not demand certainty the measurements cannot provide, and do not spend propellant correcting an error that independent evidence says is in the ground system.

ASK 23W

What is the flight director's final call, when the test that matters is whether a failure mode is still live and still fixable?

BEHIND THE BUTTON 97W

A go/no-go is not a vote on whether every uncertainty vanished. It is a decision about whether the remaining uncertainty lies inside the margins the vehicle was designed and targeted to survive.

The dangerous temptation is to react to the newest alarming number. The last tracking anomaly shared a ground timebase, and disappeared under an independent timing correction. Burning against it would create a real trajectory change to fix a measurement error.

Once entry begins, the ground loses command authority during plasma blackout. That is why the configuration, state, and onboard roles were verified before this call.

TAKES AS READ

the heat-shield configuration has been physically verified · the entry state is supported by independent tracking · onboard guidance has passed its autonomous test

VERDICT 91W

The evidence now supports entry. The shield configuration has been physically checked. The navigation solution lies inside the approved corridor, with independent support. Autonomous guidance is ready for the blackout. The late radiometric anomaly was traced to a shared ground reference rather than to spacecraft motion. An unplanned burn would add a real trajectory change without evidence that one is needed. Waiting is not a neutral option because the vehicle is already committed to atmospheric encounter. The defensible call is GO for entry under the signed checklist and predefined monitoring triggers.

TAKEAWAY 20W

The final decision is earned when the remaining uncertainty is inside verified physical margins and no unresolved actionable hazard remains.

END OF THE SHIFT 19W

Use integrated mechanics, thermodynamics, waves, fluids, uncertainty, ethics of command as an evidence chain rather than as isolated facts.

THE CLOSING CARD

The capsule vanishes into the radio blackout. Four minutes pass. Then the speaker cracks, and Hale reads the altitude. Small chutes open. Three main chutes fill above the Pacific. At 12:07, the capsule hits the water upright. All three crew members reach the recovery deck.

The return worked because you kept asking what had changed: the ship or the reading. A shared power source did not become a false cabin leak. A shared clock did not become a needless burn. You turned each risk into a margin the next call could use.

The last line under your name reads: GO for entry. Some doubt remained, but each known risk fit inside a checked limit. Three people are home tonight because you made the hard calls before the last safe choice closed.

Card text only — no options, boards or answers. 10,168 words of card prose across 52 stops. Generated from `themes/bring_them_home` by `tools/make-cardbook.mjs`.